



ENTOFARM

SUSTAINABLE AGRICULTURE AND RESOURCE MANAGEMENT

<https://www.ento.farm/>



MODULE OVERVIEW

This module introduces students to sustainable agriculture and responsible resource management. Using the example of insect farming (EntoFARMs), learners will explore how waste can be turned into valuable resources like protein for animals and fertilizer for plants.

TARGET GROUP

Age Range: 12-18 years

Subject Links: Biology, Environmental Science, STEM, Geography, Civic Education.

LEARNING OBJECTIVES

By the end of this module, students will be able to:

Knowledge: Understand sustainable agriculture and insect farming.

Skills: Identify waste reuse and suggest sustainability actions.

Attitudes: Value resource conservation and innovative food solutions.



SCAN ME

Who are we?

About the Project

This module has been developed within the framework of an **Erasmus+ proposal** focusing on **sustainable agriculture, circular economy, and innovative learning** approaches in STEM education.

The project brings together educational institutions, universities, and organizations from **Greece, Bulgaria and North Macedonia** to develop engaging educational materials for secondary school students. Its main objective is to help young people better understand how sustainable practices can address environmental challenges while **encouraging interest in STEM education and career**.

A central element of the project is the EntoFARM educational concept, which introduces students to alternative and sustainable food production systems such as insect farming and other circular resource management practices. Through these examples, students explore how biological processes, technological innovation, and responsible resource use can contribute to more sustainable food systems.

The educational materials produced in the project aim to support teachers in introducing sustainability and circular economy concepts through interactive and interdisciplinary learning activities.



Project Consortium:

- Fraud Line – Greece (Project Coordinator)
- University of Thessaly – Greece
- University of Patras – Greece
- 1st EPAL Volou – Greece
- Professional Agricultural School “Buzema” – Bulgaria
- Professional Forum for Education – Bulgaria
- Association for Internationalization of Education and Science – North Macedonia

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SUSTAINABLE AGRICULTURE

What it means:

Sustainable agriculture is farming in a way that produces enough food for people today without harming the land, water, or environment that future generations will depend on. It's like treating the Earth as a friend, not a machine. Instead of squeezing every drop of life out of the soil with heavy chemicals and over-farming, sustainable agriculture tries to keep nature in balance. Farmers use natural methods like crop rotation, composting, or using insect frass as fertilizer. This way, the soil stays fertile, water stays clean, and people still get healthy food.

Example: A farmer plants beans one year and corn the next, because beans help add nutrients back into the soil that corn takes out.



CIRCULAR ECONOMY

What it means:

A circular economy is the opposite of a "throwaway" society. In our usual system, we take resources (like wood, water, or minerals), make products, and then throw them away when we're done — this is called a linear economy. In a circular economy, we design things so that nothing becomes useless. Materials are reused, recycled, or turned into something new, again and again. In farming, this could mean food waste doesn't go to the landfill but instead is used to grow insects, which then provide protein for animals and fertilizer for plants. The idea is that the circle keeps going — waste from one process becomes the starting point for another.

Example: An orange peel from your lunch could feed insects, and the insects could then help grow more oranges in the future.



FOOD SYSTEMS

What it means:

A food system is the entire journey of food from the very beginning to the very end. It starts with how food is grown on farms, continues with how it's transported, sold, cooked, and eaten, and ends with what happens to the leftovers. It's not just farming — it includes trucks, factories, supermarkets, kitchens, and even the trash can. A food system also shows how people, animals, and the environment are connected. When food is wasted, it's not just the food that's lost — it's also the water, land, energy, and effort that went into producing it.

Example: A hamburger involves cows (farmer), grass (soil and water), transport (trucks using fuel), processing (factory), cooking (kitchen energy), and waste (leftover fries in the trash). That's all part of the food system.



RESOURCE MANAGEMENT

What it means:

Resource management is about using the Earth's natural gifts — like water, soil, forests, minerals, and energy — in a careful and responsible way. It means not using more than we need, finding ways to share resources fairly, and making sure they last for the future. Bad resource management leads to shortages, pollution, and conflicts, while good resource management means everyone gets enough without destroying the planet.

Example: Collecting rainwater for crops instead of overusing underground water that might run out.



CLIMATE CHANGE AND AGRICULTURE

What it means:

Agriculture and climate change are connected in two ways. First, farming activities can cause climate change — for example, using a lot of chemical fertilizers, raising large numbers of cattle (which produce methane gas), or cutting down forests for farmland adds greenhouse gases to the atmosphere. Second, farming suffers from the effects of climate change — droughts, floods, rising temperatures, and storms make it harder to grow food. Sustainable farming methods (like using organic fertilizers, planting trees, or reducing food waste) can help reduce emissions and make farms more resilient to extreme weather.

Example: Planting shade trees on farms helps protect crops from heat while also absorbing carbon dioxide from the air.

WASTE TO RESOURCE CYCLES

Agricultural Residues → Biochar & Soil Amendment

After harvest, farms often generate large amounts of crop residues — rice husks, corn stalks, sugarcane bagasse, or wheat straw — which are usually burned or left to rot, releasing carbon dioxide and pollutants. Instead, these residues can be transformed into biochar through pyrolysis, a process that heats organic matter in limited oxygen. The result is a stable form of carbon that stays in the soil for centuries, improving its fertility, structure, and water retention, while also helping to capture and store carbon, reducing greenhouse gas emissions.

Organic Waste → Insect Protein & Frass

This idea shows that waste doesn't have to be the "end of the line." Organic waste (like food scraps, fruit peels, or vegetable leftovers) can become the "food" for insects. These insects grow into a valuable source of protein that can feed animals like chickens or fish. At the same time, their waste (called frass) is rich in nutrients and works as a natural fertilizer for plants. This closes the loop — food waste is not wasted but transformed into something useful again.

Food Waste → Biogas & Fertilizer

Food leftovers, coffee grounds, and spoiled produce can be placed in anaerobic digesters — systems that use microorganisms to break down organic matter without oxygen. This process generates biogas (a renewable energy source for heating or electricity) and digestate, a nutrient-rich residue used as a biofertilizer.

Used Cooking Oil → Biodiesel

Restaurants, school cafeterias, and households generate large amounts of used cooking oil, which can cause pollution if poured down the drain. Instead, this oil can be collected and processed through transesterification, a chemical reaction that converts fats and oils into biodiesel — a renewable and cleaner-burning fuel. The process also produces glycerin, which can be used in soaps and cosmetics, adding another layer of resource recovery.

SHORT DEFINITIONS

Sustainable agriculture



Farming that cares for
the future

Circular economy



No waste, only reuse

Food systems



The whole food
journey from farm to
fork to waste

Resource management



Using nature's gifts
wisely

Climate change & agriculture



Farming needs to fight
and adapt to global
warming

Waste-to-resource cycles



Turning trash into
treasure

REAL WORLD APPLICATIONS

1. REDUCING FOOD WASTE AT SCHOOL AND HOME

What it means:

Food waste happens when edible food is discarded instead of being eaten. It can result from oversized portions, imperfect produce, or poor storage. Every bit of wasted food represents wasted resources — water, energy, soil nutrients, and human labor. Reducing food waste is one of the easiest ways individuals and communities can combat climate change and promote sustainability.

How it looks in practice:



At school:

Students can design a “Smart Plate Challenge”, encouraging peers to take only what they can eat. Cafeterias can create a “share station” where unopened food or whole fruits are placed for others to take. Compost bins near dining areas can collect unavoidable waste for the school garden.



At home:

Families can organize weekly meal plans and keep a “use first” box in the fridge for food nearing expiration. Leftovers can be turned into creative dishes — for example, mashed vegetables become soup, and stale bread becomes French toast or breadcrumbs.



In the community:

Local shops and bakeries can partner with charities to donate unsold food instead of discarding it, building a food recovery network that feeds people and reduces waste.

Why it matters:

Globally, around one-third of all food produced is lost or wasted, contributing to nearly 8–10% of greenhouse gas emissions. If food waste were a country, it would be the third largest emitter after the U.S. and China. Reducing waste saves money, nourishes communities, and protects the planet.

REAL WORLD APPLICATIONS

2. USING INSECT PROTEIN AS ANIMAL FEED OR SOIL ENHANCER

What it means:

Insects such as black soldier flies, mealworms, and crickets can transform organic waste into high-value protein for livestock and fish. This sustainable protein source reduces the need for soy and fishmeal — both major drivers of deforestation and overfishing. The insects' by-product, frass, is rich in nitrogen, phosphorus, and potassium, making it a perfect organic fertilizer.

How it looks in practice:



At school:

Students can run a small EntoFARM project, feeding food scraps from the cafeteria to black soldier fly larvae. After harvesting, the dried larvae can be donated to local farmers for chicken or fish feed.



At home:

The frass produced is mixed with soil to improve plant growth naturally, replacing chemical fertilizers.



In the community:

City councils can establish insect-based recycling centers where neighborhood food waste is converted into animal feed and fertilizers, closing the urban food loop.



In the industry:

Aquaculture companies are beginning to replace traditional fishmeal with insect meal, reducing costs and pressure on wild fish stocks.

Why it matters:

Insect protein is eco-efficient, requiring far less land, water, and energy than conventional livestock feed. It provides a practical path toward circular agriculture, turning waste into nutrition and soil health while reducing environmental impact.

REAL WORLD APPLICATIONS

3. WATER AND SOIL CONSERVATION TECHNIQUES

What it means:

Water and soil are the lifeblood of agriculture. Misuse — through over-irrigation, chemical overuse, or erosion — threatens both food security and ecosystems. Conservation means protecting these resources through smarter techniques that keep soil fertile and water clean for future generations.

How it looks in practice:



At school:

Students can install a rainwater harvesting system with barrels connected to gutters, using stored water to irrigate the school garden. Science classes can monitor soil moisture with sensors to study water efficiency.



In farming:

Drip irrigation delivers water directly to roots, reducing evaporation and waste. Mulching and cover crops prevent erosion and help the soil retain moisture. Farmers can also rotate crops and apply organic compost or biochar to maintain fertility.



At home:

Turning off taps, reusing greywater for plants, and fixing leaks make a real difference. Even planting native species in gardens reduces irrigation needs.

Why it matters:

Globally, agriculture uses about 70% of all freshwater withdrawals. Soil degradation affects nearly one-third of Earth's land. Without healthy soil and water, farming becomes unsustainable, threatening food production and livelihoods. Conservation ensures these vital resources stay productive and resilient under changing climates.

IN SHORT

Reducing food waste



Saves money,
resources, and cuts
emissions.

Insect protein & frass



Transforms waste into
feed and fertilizer.

Water & soil conservation



Protects the
foundation of life and
farming.

CASE STUDY: SCHOOLS FOOD WEB – COMPOSTING IN UK SCHOOLS

What was done?

- The “Schools Food Web Programme” helped schools audit their food systems (what’s wasted, what’s eaten) and introduced composting of unavoidable food waste.
- Schools received resources (bins, wormeries, teaching materials) and workshops.
- The compost produced was used in school gardens.



Location / Context:

20 schools in Shropshire & Telford & Wrekin, UK



Outcomes / Lessons:

Food waste was reduced and valorized (turned into compost).

The program linked waste reduction, gardening, and curriculum learning (biology, ecology).

It showed that with training and support, schools can integrate waste management into daily operations.

CASE STUDY: INSECT FARMING / BLACK SOLDIER FLY (BSF) IN KENYA

What was done?

- Farmers and entrepreneurs set up Black Soldier Fly (BSF) farms: feeding organic waste to larvae, harvesting larvae as protein feed, and using the leftover frass as fertilizer.
- Studies were done to determine which factors lead to profitability (e.g. substrate amount, labor input).
- Feasibility studies by companies (like Finland-based Manna Insect) confirmed that insect-based circular economy solutions (biowaste → insect feed + fertilizer) work well in Kenyan climate.



Location / Context:

Kenya (several other counties, smallholder farmers)



Outcomes / Lessons:

The BSF enterprise is profitable under the right conditions (efficient substrate use, controlled labor)

It helps reduce reliance on expensive feed imports and provides organic fertilizer locally.

Challenges include scaling, managing operations, and training farmers.

CASE STUDY: **IRRIGATION** **MODERNIZATION (DRIP** **IRRIGATION) IN SPAIN**

What was done?

- Spain invested heavily in modernizing irrigation systems—converting many farms from traditional flood or gravity irrigation to drip and other efficient systems.
- Studies compare water and energy use before and after modernization.
- In olive farming, use of subsurface drip lines reduced water use significantly per hectare (from around 1,800 m³/ha to ~650 m³/ha) while improving oil quality.



Location / Context:

Spain, especially between 2002 and 2015 during irrigation modernization



Outcomes / Lessons:

Substantial water savings were achieved across many farms.

Some unintended effects: as water became more available, farmers sometimes expanded cultivated area or used more water-demanding crops, which can reduce overall water gains if not managed properly.

Institutional and governance changes were needed (water rights, collective organizations) to support efficient use.

CASE STUDY: SCHOOL COMPOSTING AS LEARNING LABORATORY (BERN, SWITZERLAND)

What was done?

- A school class collaborated with a project team to set up a composting system at the school.
- The project aimed to reduce waste, lower CO₂ emissions, teach circular economy principles, and engage the community.
- It integrated composting into lessons and encouraged behavior change beyond the school.



Location / Context:

Länggass neighborhood, Bern,
Switzerland



Outcomes / Lessons:

The school became a “learning laboratory” where students can see sustainability in action.

It connects classroom theory to real-world impact (less waste, soil improvement, awareness).

Shows the value of linking multiple stakeholders (students, school administration, community) in sustainability projects.

ASSESSMENT & EVALUATION

Quiz: Sustainable Agriculture & Resource Management

Part A – Multiple Choice

What does sustainable agriculture mean?

- a) Farming with as many chemicals as possible
- b) Farming that protects the environment and future generations
- c) Farming only in big cities
- d) Farming that uses no water

What is a circular economy?

- a) An economy where waste is reused and recycled
- b) An economy where everything is thrown away
- c) Farming in circles
- d) A new kind of money

Which of these is not an example of food waste?

- a) Throwing away a half-eaten sandwich
- b) Composting vegetable scraps
- c) Not finishing school lunch
- d) Discarding “ugly” fruit

Frass is:

- a) A type of insect protein
- b) Insect waste used as fertilizer
- c) A harmful chemical
- d) A kind of soil

Which farming method saves the most water?

- a) Flooding fields with water
- b) Drip irrigation
- c) Leaving fields dry
- d) Spraying water randomly

What happens when food is wasted?

- a) Only money is lost
- b) Natural resources like water and energy are wasted too
- c) The food disappears harmlessly
- d) It makes soil healthier

Why are insects considered a sustainable protein source?

- a) They are cheap to produce and reduce food waste
- b) They require large amounts of land and water
- c) They increase pollution
- d) They replace vegetables in our diet

What is one main benefit of using biochar in soil?

- a) It makes the soil dry faster
- b) It increases soil fertility and stores carbon
- c) It replaces fertilizers made from insects
- d) It harms plant roots

ASSESSMENT & EVALUATION

Quiz: Sustainable Agriculture & Resource Management

Part B – True or False

- Insects used in farming are always dirty and unsafe.
False
- Agriculture both affects climate change and is affected by it.
True
- Using leftovers creatively (like turning old bread into croutons) is an example of reducing food waste.
True
- A circular economy means using things only once.
False
- Healthy soil is essential for strong food systems and human health.
True

Part C – Reflection & Application

1. Think of one example from your school or home where waste could become a resource. How would you redesign it using circular economy principles?
2. Why is conserving soil and water important for your community's future?
3. If your school started a mini sustainability project, what role would you like to play (planner, designer, communicator, grower, or innovator)?

ADDITIONAL RESOURCES

Links:

- [youtube.com/watch?time_continue=16&v=mEzPmb1hQ8Q&embeds_referring_euri=https%3A%2F%2Fchatgpt.com%2F&source_ve_path=Mjg2NjY](https://www.youtube.com/watch?time_continue=16&v=mEzPmb1hQ8Q&embeds_referring_euri=https%3A%2F%2Fchatgpt.com%2F&source_ve_path=Mjg2NjY)
- <https://www.youtube.com/watch?v=divmWrMGkHo>
- <https://www.youtube.com/watch?v=ilqi3Fy2kXM>
- <https://www.youtube.com/watch?v=iloAQmroRK0&list=PLWltQ6Oy0zpgxVhd2vZqTDvVXpPhSVd0>
- <https://www.youtube.com/watch?v=IXmtmeA3Ld0>
- <https://www.youtube.com/watch?v=ZERrpFwETgs>
- <https://www.youtube.com/watch?v=HVRtgH5rIAU>

TEACHER REFLECTION AND ADAPTATION NOTES

This module has been designed as a flexible educational resource that can be adapted to different classroom contexts, student backgrounds, and educational levels.

Teachers are encouraged to reflect on how the activities and content can be adjusted to best support the needs of their students.

Reflection Questions for Teachers

Before or after implementing the module, teachers may consider the following questions:

- Did the activities engage students with different interests and learning styles?
- Were all students encouraged to participate actively in discussions and group work?
- Did the module help students understand the connection between science, technology, and sustainability?
- Were examples and explanations accessible to students with different levels of prior knowledge?

Accessibility Considerations

To ensure that the module is accessible to all learners, teachers may consider:

- simplifying technical terminology when necessary
- providing visual aids, diagrams, or short videos
- allowing students to express their understanding through different formats (presentations, posters, discussions, or written reflections)
- offering additional guidance or scaffolding for students who may need more support

Supporting Girls' Engagement in STEM

Research shows that girls are more likely to engage with STEM when learning environments are collaborative and when scientific topics are linked to real-world challenges and social impact.

Teachers can therefore:

- encourage mixed-gender teamwork and equal participation
- highlight female role models in sustainability science and agriculture
- frame STEM challenges as problem-solving activities that benefit society and the environment
- ensure that classroom discussions give equal voice to all students

TEACHER REFLECTION AND ADAPTATION NOTES

Adapting the Module

Teachers are free to adapt the module by:

- adjusting the depth of scientific explanations
- modifying activities to fit available time or resources
- connecting the topic to local environmental or agricultural examples
- integrating the module with subjects such as biology, geography, environmental science, or technology

The goal is to ensure that the module remains relevant, inclusive, and meaningful for students in different educational settings.

Do you have any questions?

Please let us know! We are happy to help.

Contact us at:



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Enriching lives, opening minds.